JaeHyun Lee

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RESEARCH INTERESTS

Computer Graphics, Physics-Based Animation/Simulation, PDE, High-performance Computing, Numerical methods, Optimization

EDUCATION

University of Wisconsin-Madison — Wisconsin, USA
Ph.D. in Computer Sciences — *Sep. 2024 – present*

- Advisor: Prof. Eftychios Sifakis

Korea University — Seoul, Republic of Korea
M.S. in Computer Science and Engineering — *Sep. 2021 – Feb. 2024*

- Advisor: Prof. JungHyun Han and Prof. Kiwon Um
- GPA: 3.93/4.0

Korea University — Seoul, Republic of Korea
B.S. in Computer Science and Engineering (Double major) — *Mar. 2019 – Feb. 2021*
B.S. in Mechanical Engineering — *Mar. 2015 – Feb. 2021*

- Including 2 years of military service
- GPA: 3.98/4.0
- **Graduated with Great Honor *(Summa Cum Laude)***

PUBLICATIONS

Machine Learning Operator Lexicon for Partial Differential Equation Solvers on Irregular and Sparse Domains.
SIGGRAPH Asia 2026 (To appear).
JaeHyun Lee and Eftychios Sifakis.

Dimension Expansion for Untangling Mass-Spring System.
Computer Animation and Virtual Worlds (CAVW) 2025. [paper] [video]
Seung-wook Kim, HuiSeong Lee, JaeHyun Lee, Kiwon Um, and JungHyun Han.

Momentum-preserving inversion alleviation for elastic material simulation.
Computer Animation and Social Agents (CASA) 2024. [paper] [video]
Heejo Jeong, Seung-wook Kim, JaeHyun Lee, Kiwon Um, Min Hyung Kee, and JungHyun Han.

Inversion alleviation for stable elastic body simulation.
Computer Animation and Social Agents (CASA) 2023. [paper] [video]
JaeHyun Lee, Seung-wook Kim, Kiwon Um, Min Hyung Kee, and JungHyun Han.

EXPERIENCE

NVIDIA High-Fidelity Physics Research Lab — Santa Clara
Researcher Intern — *May 2026 – Present*

PROJECTS

General Purpose Material Point Method (MPM) Engine — Korea University
Researcher — *Oct. 2023 – May. 2024*

- Developed C++, CUDA-based state-of-the-art MPM framework, with visualization system using OpenGL. [code]

LG Electronics: Air Conditioning Airflow Simulation Visualization System — Korea University
Project Assistant — *Mar. 2022 – Aug. 2022*

- Contributed to the project by implementing Python-based, GPU-accelerated real-time airflow simulator visualized with volume rendering. The project won the **first prize** among 489 teams. [code] [video]

Collision Detection for Constrained Projective Dynamics (CPD) — Korea University
Researcher — *Dec. 2020 – May. 2021*

- Implemented tetrahedral collision detection module for ACM Transactions on Graphics 2021 paper titled 'Constrained Projective Dynamics: Real-Time Simulation of Deformable Objects with Energy-Momentum Conservation'. [paper] [video] [code]

TEACHING

Parallel & Throughput - Optimized Programming (CS557) — University of Wisconsin-Madison
Teaching Assistant — *Spring 2026*

Computer Graphics (CS559) — University of Wisconsin-Madison
Teaching Assistant — *Spring 2025, Fall 2025*

Programming III (CS400) — University of Wisconsin-Madison
Teaching Assistant — *Fall 2024*

Computer Graphics (COSE331) — Korea University
Teaching Assistant — *Spring 2022*

SCHOLARSHIPS

Kwanjeong Educational Foundation Scholarship, Kwanjeong Educational Foundation — *Spring 2022 – Fall 2023*

Teaching Assistant Scholarship, Korea University — *Spring 2022*

Research Scholarships, Korea University — *Fall 2021, Fall 2022*

National Science and Engineering Scholarship, Ministry of Science and ICT — *Spring 2019 – Fall 2020*

Special Scholarships, Korea University — *Spring, Fall 2018*

HONORS AND AWARDS

Best Industry-Academic Project Award, Ministry of Trade, Industry and Energy — *Nov 2023*

Best Research award, Korea Electronics Association — *Feb 2022, Dec 2022, Aug 2023*

Great Honor, Korea University — *Graduation*

President's List, Korea University — *Fall 2018 – Spring 2019*

Dean's List, Korea University — *Spring 2018*

Semester High Honors, Korea University — *Spring 2017 – Spring 2020*

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript

APIs: OpenGL, CUDA, Three.JS ,OpenMP

Other Tools and Libraries: Git, Eigen, Partio, ImGui, Assimp, PyTorch, Fusion360, CMake, Taichi Lang, Blender

LANGUAGE LEVEL

Korean: Native

English: Fluent